AI-Based Programming Course

Project Deliverables & Requirements

This document outlines the required deliverables for the course project. The objective is to build a complete AI pipeline starting from raw data all the way to deployment, while demonstrating understanding of core concepts covered throughout the course.

1. Dataset Requirement

- The dataset must be unstructured data, such as: Images, Text, and Audio
- **The dataset should be:**
- Clearly described (source, size, classes, etc.)
- Properly organized and documented

2. Data Preprocessing

Apply suitable preprocessing techniques depending on data type

Deliverable:

- Explanation of preprocessing steps
- Code implementation
- Before vs After examples

3. Feature Selection

Apply and compare: **Heuristic-based feature selection** and **Genetic Algorithm (GA) feature selection**

Requirements:

- Apply both techniques on the same dataset
- Compare based on:
- Performance (Accuracy / F1-score)
- Number of selected features

Deliverable:

- Clear explanation of both methods
- Comparison table + analysis

4. Ensemble Learning

Implement and compare: **Voting Classifier** and **Stacking Classifier**

Requirements:

- Use **at least 3 base models**
- Evaluate performance of: **Individual models**, **Voting**, and **Stacking**

Deliverable:

- Performance comparison table
- Discussion of results

5. Explainable AI (XAI)

Apply: **LIME and SHAP**

Requirements:

- Explain model predictions

Provide:

- Local explanation (specific prediction)
- Global explanation (feature importance)

Deliverable:

- Visualization outputs
- Interpretation of results

6. Graphical User Interface (GUI)

Design a user interface for the system.

Team Size vs Expected Quality:

- **5 or more** students → **High-quality, professional GUI** (Required)
- **3–4** students → **Medium-quality GUI** (Acceptable)
- **2** students → **Basic GUI** (Minimum requirement)

Requirements:

- User input
- Prediction display
- Explanation display (from XAI)

7. Deployment

Deploy the system as a working application.

Requirements:

- Use a framework such as:
- Flask
- Provide:
- API endpoint (/predict)
- Input → Output workflow
- Logging
- Model versioning

Final Submission Must Include

1. Source Code
2. Report (explaining all steps)
3. Results & Comparisons
4. GUI Demo
5. Deployment Demo
6. Evaluation Criteria

Projects will be evaluated based on:

1. Correct implementation of all stages
2. Quality of analysis and comparison
3. Code clarity and organization
4. GUI quality (based on team size)
5. Completeness of deployment

Important Notes

1. All steps must be implemented and justified
2. Comparisons are mandatory (not optional)
3. Copying or using ready-made pipelines without understanding will result in penalties